MILITARY SPECIFICATION

RADIACMETERS, QUARTZ-FIBER SELF-INDICATING
GENERAL SPECIFICATION

This amendment forms a part of Military Specification MIL-R-24037A, dated 12 December 1969, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 2

* 3.1, Delete and substitute.

"3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.2).".

Page 3

3.7.4, line 2: Add "Plastic end caps which have been tested and specifically approved by the procuring activity may be used on the barrel of the radiacmeter. Plastic end caps shall fit over the metal barrel, shall be permanently attached to the barrel, and shall not interfere with reading or charging of the radiacmeter. Radiacmeter barrel with plastic end caps attached shall not exceed 1.7 centimeters (cm) (0.6875 (11/16) inch (in.)) in diameter.".

3.7.4.1, line 4: Delete "PP-4726( )/PD" and substitute "PP-4276( )/PD".

3.7.6, first sentence: Delete and substitute "Unless otherwise specified (see 6.1.1), the radiacmeters shall be provided with a double jaw alligator type metal clip with a coil spring spot welded (two points) or riveted (one or two rivets) to a collar no less than 0.64 cm (0.25 (1/4) in.) in width. A prototype clip assembly for clips which are attached by rivets shall be approved by the procuring activity prior to use.

Page 4

3.8.1, line 3: Delete "250 volts" and substitute "190 volts"; line 4: Delete "200 volts" and substitute "250 volts".

Add the following new paragraph:

"3.8.7 Fiber and wall conduction. The fiber and the walls of the ionization chamber of the radiacmeter shall be sufficiently conducting so that the fiber will maintain its position within a ±5 percent of full scale when the polarity of charge on the fiber is rapidly changed from positive to negative or negative to positive. The testing shall be conducted in accordance with 4.6.26.".

Page 5

3.9, Item (c): Delete "50,000" and substitute "100,000"; Item (e): Delete "4.6.21" and substitute "4.6.20".

3.10, Item (b): Delete "virbation" and substitute "vibration".

FSC 6665

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* 4.2, Item (a): Delete and substitute:

"(a) First article inspection (see 4.3)".

* 4.2, Item (c): Delete.

* 4.3, Delete and substitute:

"4.3 First article inspection. Unless otherwise specified (see 6.1.1), first article inspection is required on each type dosimeter specified in the detail specifications. Twenty (20) dosimeters of a given type constitute a sample for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE I, and the detail specifications.".

* TABLE I: Delete "title" and substitute "First article inspection.".

TABLE I, add the following at end of table:

* Fiber and wall conduction	3.8.7	4.6.26	".

* 4.3.1 and 4.3.2: Delete in their entirety.

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* 4.3.2.1, 4.3.2.2, 4.3.2.3, 4.3.2.4, 4.4, 4.4.1, and 4.4.2: Delete in their entirety.

Page 8

4.5.1.1.4, first sentence: Delete and substitute "A sample number of radiacmeters in accordance with TABLE II shall be selected from the end of each month's production and shall be subjected to each of the samples' respective Group C, subgroup tests specified in 4.5.1.4.".

TABLE III, add the following at the end of Sub-group II:

* Fiber and wall conduction	3.8.7	4.6.26	".

Page 11

4.6.11, line 2: Delete "(kV)" and substitute "(keV)".

Page 12

4.6.24, line 1: Delete "Kiethly" and substitute "Keithley".

Add the following new paragraph:

"4.6.26 Fiber and wall conduction test. The radiacmeter shall be tested to ensure conformance to the requirements of 3.8.7 with a dosimeter charger having the following characteristics:

a. A switch which can reverse the polarity of the DC output voltage to the charging pedestal with a maximum delay time of less than 0.2 second.
b. A continuously variable voltage output over a range of 0 to ±250 VDC with a precision of ±2.0 VDC.
c. For a given voltage output of either polarity, when the polarity is changed, the reversed voltage output shall be within ±1.0 VDC.

The radiacmeter shall be charged with a positive polarity to provide a fiber image setting of 10 percent of the full scale reading. The radiacmeter shall be stored at ambient conditions for a period of 24 hours. At the end of the 24-hour storage period, the radiacmeter shall be reset at 10 percent of full scale with a positive polarity. While the radiacmeter charging switch is closed at this setting, the polarity switch shall be changed from a positive to a negative polarity. The fiber image of the radiacmeter shall return to the 10 percent of full scale setting within ±5 percent of full scale. The time it takes the fiber image to return to the 10 percent of full scale setting within ±5 percent of full scale shall not exceed 1.0 second after the polarity switch is changed. The fiber image shall come to rest in a fixed position within these limits and shall do so within 2.0 seconds after the polarity is changed."

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* 6.1.1, after "(g)", add the following new item:

"(h) Number of first article samples to be submitted if other than specified in 4.3".

* 6.1.2, Item (b), delete and substitute:

"(b) First article test date (see 4.3).".

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* 6.2 and 6.3, Delete and substitute:

"6.2 First article. When a first article is required it should be tested and approved under the appropriate provisions of 52.209-3 of the Federal Acquisition Regulations. The first article should be a first production item. The first article should consist of 20 dosimeters of a given type. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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